UC San Diego
RADY SCHOOL OF MANAGEMENT

MGT 413: Operations, Information Systems and Data Analysis

Winter 2026

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DESCRIPTION

Operations Management (OM) involves the systematic design, execution, and improvement of business processes, projects, and partner relationships. Effective management of operations is vital to every type of organization in today's context, when we see that significant competitive advantages accrue to those firms that can excel in execution and achieve sustained and profitable growth. This course goes beyond cost minimization and is an introduction to the core set of issues that firms large and small must confront and coordinate in their journey towards sustained scalability, growth and profitability.

OBJECTIVES

At the close of this course, you will be able to:

- Understand Operations as a process
- Be able to link the strategy with the process
- Apply Operational metrics in the real world
- Synthesize information about process and analyze the relationship among metrics
- Choose the best methods for evaluating and improving the process
- Understand the role of inventory in the real world and how to analyze in operations perspective
- Apply Lean Operations concepts in the real world
- Better understand how to mitigate the risks in the process via inventory/capacity/flexibility

MATERIALS

Required

- Managing Business Process Flows: Principles of Operations Management by Anupindi et al., 3nd Edition, 2012, Prentice Hall, ISBN: 0136036376.
- The Goal: A Process of Ongoing Improvement by Eliyahu Goldratt, 2004, 3rd Edition, North River Press, ISBN: 0884271781.

Recommended

- Matching Supply with Demand: An Introduction to Operations Management by Gérard Cachon and Christian Terwiesch.

SCHEDULE

CP is the coursepack with cases, wcan be found in Study.Net Materials.
MBPF is the textbook, Managing Business Process Flows, can be found in BryteWave.

Date	Assignments	Class Topic & Activities	Readings
Week 1		Warm up & Motivating Examples; Strategic Framework	Shouldice case in CP Deep Change in CP Chapters 1 & 2 of MBPF
Week 2	HW #1	Process Flow Analysis; Types of Processes; Operations Metrics; Little's Law;	Learning by the case method in CP State Automobile License Renewal case in CP Chapters 3 & 4 of MBPF
Week 3		Capacity Analysis; Inventory build-up Analysis	Capacity in CP (and Fishing Fleet & Canary problem) Chapter 5 in MBPF
Week 4	National Cranberry Coop case write-up	Product Mix Selling; Theoretical Flow Time & Critical Path; National Cranberry case; Littlefield Technologies	Chapter 5 in MBPF; National Cranberry case; Littlefield Technologies (LT) Overview; Capacity Management at LT;
Week 5	LT round 1 simulation game (begins Jan 30); HW #2	House Build Game; Guest Lecture: Jimmy Anklesaria	Review the National Cranberry Case
Week 6	Quiz #1	Inventory Analysis; EOQ, ROP; Safety Stock; Review the LT game round 1	Radical Waves DB in CP; Chapter 6 in MBPF;
Week 7	HW #3 Quiz #2	C_u, C_o, Newsvendor Analysis; Optimal Service Level; Inventory Pooling;	Chapter 7 in MBPF; Sirens Coffee Beans in CP;
Week 8	Quiz #3	Safety Capacity; Queueing Principles;	There's more to a Line than its wait in CP

		Inventory and Variability	Note on Queueing in CP Chapter 8 in MBPF
Week 9	LT round 2 simulation game	Beer Game; Supply Chain Management	The bullwhip effect in Supply Chains in CP
Week 10	LT round 2 team report	Lean Operations	Toyota Motor Manufacturing Chapter 10 in MBPF Benihana of Tokyo
Week 11	Final Exam	Review	

ASSIGNMENTS: C – Collaboration with classmates is allowed. However, each student must submit.

There will be three homework assignments spread throughout the quarter. Each assignment will be posted on Canvas with a due date and is due *at the beginning* of class on its due date, as we may cover related material during the class session. Each homework question allows up to two attempts; the higher score will be used as the final grade. Although you may work with your study group on homework assignments, all submitted work must be your own and completed individually.

Reading Assignments: No submission of report is required for reading assignments.

For each class session, I will post a set of guiding questions, discussion points, and goals for the lecture on Canvas. You are expected to prepare answers to these questions and be able to provide substantiated arguments on discussion points. Feel free to work with others to prepare for these class assignments. I suggest that you work closely with your study groups on these class assignments to share and discuss individual ideas so that you come prepared to class and contribute to a richer learning experience. It is required that you read the assigned readings before coming to class. This will help give you perspective on the topics to be covered.

Team Projects I (National Cranberry Case Write-up): G – Students may work together in groups and turn in one project or assignment for the entire group.

There will be one case write-up worth a total of 20 points. One report per team should be handed in at the beginning of class on the day we cover each case. The goal of this exercise is for you to discuss the case in detail with your teams and then together agree upon some recommendations. These recommendations should be backed up by solid arguments, evidence, and perhaps even mathematical analysis in some instances. The case designated for write-up is National Cranberry Case. You will have a better idea of what I expect in a case analysis by observing how we cover the case and other analysis in sessions 2 and 3. The guiding questions for the case write-up will be posted in Canvas.

Team Projects II (Littlefield Technologies): G – Students may work together in groups and turn in one project or assignment for the entire group.

You will play an Operations simulation game, called Littlefield Technologies. There will be two separate rounds and each round runs for one week. More details about the Littlefield Technologies will be discussed, and given in the coursepack as well as in Canvas.

For the first Littlefield Technologies simulation (Capacity scenario**), it starts at 6 PM PST on January 30th, and ends on February 6th at 6PM PST.** The grading scheme is as follows (4 pts max):

• 4 pts – Top two teams
• 3 pts – Top half of remaining teams OR earnings greater than $1,200,000
• 2 pt – Rest of remaining teams

For the second Littlefield Technologies simulation (Forecasting, Capacity, and Inventory scenario), **it starts at 6PM PST Feb 27th, and ends on March 6th, at 6PM PST.** The grading scheme is as follows
(6 pts max):

• 6 pts – Top two teams
• 5 pts – Rank 3 – 5 OR earnings greater than $2,200,000
• 4 pts – Rank 6 – 10 OR earnings greater than $2,000,000
• 2 pt – Rest of remaining teams

Following the second Littlefield Technologies simulation, each team will write-up a report. The grading of this report is independent of how your team performed in the simulation. (10 pts max)

Quizzes: I – Independent, individual work only. No collaboration or consultation allowed. We will have quizzes sporadically across the quarter. Each quiz will be posted on Canvas. Each quiz allows up to two attempts; the higher score will count toward the final grade.

Exams: I – Independent, individual work only. No collaboration or consultation allowed.
We have the final exam. It will be three-hour long and open book exam.

Class Participation

Your class participation grade is based upon your contribution to the discussions. To prepare for class, you should use the class assignments merely as a starting point. At the beginning of the class with a case, one or more class members will be asked to start the session by addressing a specific question. Anyone who has prepared the case or answers to the class assignment will have no problem handling such a request. After a few minutes of initial analysis and recommendations, we will open the discussion to the rest of the class. As a group, we will try to build a complete analysis of the situation. You are expected to be an active participant throughout the entire class and to contribute to the quality of the discussion. Please note that the frequency of your interventions in class is not a key criterion for effective class contribution. Some criteria used to evaluate class contribution are as follows:

1. Is the participant a good listener?
2. Are the points that are made relevant to the discussion? Are they linked to the comments of others?
3. Do the comments show evidence of analysis of the case?
4. Is there willingness to test new ideas, or are all comments “safe”? (For example, repetition of case facts without analysis and conclusions.)
5. Do comments clarify or build upon the important aspects of earlier comments and lead to a clearer understanding of the case?

GRADING

Assignments	Points [or percentage]
Class Participation	10%
Littlefield Technologies	20%
Quizzes	10%
Case Write-up	20%
Homework Assignments	10%
Final Exam	30%
Total	100

COURSE POLICIES

You can work with anyone in the class on Class Assignments. You should work mostly with your assigned study group on Homework Assignments. However, you must individually write-up your own solution to be submitted. **For the two Littlefield Technologies Assignments, you should limit discussion *only* to your study group since these are team competitions.** For the Final Exam, you are on your own. Therefore, I suggest that you work individually enough to be well prepared for the exams. For example, solving each homework assignment individually and then discussing it with your study group might be a good strategy.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2

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STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or **osd@ucsd.edu.**